

# Business Requirements Document (BRD)

Production Bounded Conversational RAG System for 1:1 Global Academic Tutoring

| DOCUMENT STATUS     | CURRENT VERSION            | TARGET DOMAIN        | RELEASE DATE       |
|---------------------|----------------------------|----------------------|--------------------|
| Production Baseline | v1.0.0 (Release Certified) | wementors.vercel.app | September 23, 2026 |

|                                                                           |                                                                        |                                                                                |                                                                                |
|---------------------------------------------------------------------------|------------------------------------------------------------------------|--------------------------------------------------------------------------------|--------------------------------------------------------------------------------|
| <b>ANSWER CORRECTNESS</b><br><b>88.1%</b><br>+24.6 pp vs Baseline (63.5%) | <b>TRANSACTION SAFETY</b><br><b>100.0%</b><br>Zero Fake Booking Claims | <b>BENCHMARK HALLUCINATION</b><br><b>0.0%</b><br>126 Evaluated Benchmark Cases | <b>RELEASE VERIFICATION</b><br><b>20 / 20</b><br>1,146 Automated Checks Passed |
|---------------------------------------------------------------------------|------------------------------------------------------------------------|--------------------------------------------------------------------------------|--------------------------------------------------------------------------------|

## Document Control & Sign-off Hierarchy

| Stakeholder Role   | Name & Title                         | Department              | Approval | Date         |
|--------------------|--------------------------------------|-------------------------|----------|--------------|
| Executive Sponsor  | Dr. A. Ramanathan, Managing Director | Executive Board         | APPROVED | Sep 21, 2026 |
| Head of Academics  | S. Mukherjee, Academic Dean          | Curriculum & Pedagogy   | APPROVED | Sep 21, 2026 |
| Lead AI Architect  | R. K. Varma, Principal Engineer      | Platform Engineering    | APPROVED | Sep 22, 2026 |
| Head of Admissions | P. Nair, Enrollment Director         | Global Counseling & CRM | APPROVED | Sep 22, 2026 |

## 1. Executive Summary

**WeMentors Online Academy** delivers personalized 1:1 online academic mentoring for K-12 students (Grades 3 through 10) across four core international curricula: **CBSE, ICSE, Cambridge (IGCSE), and International Baccalaureate (IB)**, complemented by its specialized communicative development program, **The Confident Speaker**.

**The Business Problem:** Prospective parents evaluating educational programs represent high-value, high-consideration traffic. Historically, static website FAQs failed to satisfy multi-turn parent queries regarding tutor compatibility, board curricula, and trial classes, while off-the-shelf generative AI chatbots posed unacceptable corporate risks by fabricating unauthorized fee discounts and hallucinating synthetic booking confirmations.

**The Solution:** The WeMentors Conversational AI Assistant delivers a bounded, deterministic-first conversational retrieval pipeline. By replacing unconstrained generation with a **Weighted Lexical-Semantic Retriever** across 41 canonical knowledge items and enforcing a **3-Tier Multi-Provider Fallback Engine** (Google Gemini 3.5 Flash-Lite → Groq Cloud LLMs → Zero-Network Offline KB), the system guarantees 100% truthful academic representation and zero phantom bookings.

## 2. Project Objectives & Key Performance Indicators

The primary commercial objective is converting evaluating website visitors into scheduled Free Demo Class leads without creating customer expectation mismatches or human counselor burnout.

| Metric Dimension            | Baseline (Unguided) | Target Threshold | Production v1.0 Result   | Business Impact                                             |
|-----------------------------|---------------------|------------------|--------------------------|-------------------------------------------------------------|
| Answer Correctness          | 63.50%              | ≥ 85.00%         | <b>88.10%</b> (+24.6 pp) | Parents receive verified, high-confidence guidance.         |
| Demo Transaction Safety     | 100.00%             | 100.00%          | <b>100.00%</b> (0 leaks) | Zero fake appointments; 100% routed to verified form.       |
| Benchmark Hallucination     | 14.20%              | ≤ 2.00%          | <b>0.00%</b> (126 cases) | Zero invented policies, fee discounts, or fake boards.      |
| Context Resolution Accuracy | 92.10%              | ≥ 90.00%         | <b>96.83%</b>            | Fluid pronoun and ordinal handling in multi-turn dialogues. |
| Provider Failover Recovery  | Failed (Crash)      | <1,000 ms        | <b>588.3 ms</b>          | Uninterrupted counseling during upstream API rate limits.   |

## 3. Project Scope & Boundary Definition

|                                                                                       |         |
|---------------------------------------------------------------------------------------|---------|
| +-----+<br>  PROJECT SCOPE MATRIX  <br>+-----+                                        |         |
| [IN-SCOPE: v1.0 Production Baseline]                                                  |         |
| * 1:1 Academic Mentoring for Grades 3-10 across CBSE, ICSE, Cambridge (IGCSE), and IB |         |
| * Confident Speaker communicative & public speaking program advisory                  |         |
| * Multi-turn conversational memory with pronoun and anaphoric state resolution        |         |
| * Weighted lexical-semantic TF-IDF retrieval across 41 canonical knowledge items      |         |
| * 3-Tier Multi-Provider Fallback (Gemini Flash-Lite → Groq LLM → Offline KB)          |         |
| * Strict Demo Transaction Refusal Contract (redirect to official web form)            |         |
| * Responsive frontend widget with dark/light themes and quick suggestion chips        |         |
| +-----+                                                                               | +-----+ |
| [OUT-OF-SCOPE: v1.0 Production Baseline]                                              |         |
| * Collecting credit card payments or executing financial transactions inside chat     |         |
| * Autonomous calendar booking without routing through the official intake form        |         |
| * Homework solving, student grading, or generating examination answer keys            |         |
| * Medical, psychological, or non-educational advice                                   |         |
| +-----+                                                                               | +-----+ |
| [FUTURE PHASES: Roadmap]                                                              |         |
| * Phase 1.5: Direct CRM webhook integration for verified in-chat lead submission      |         |
| * Phase 2.0: Multi-lingual localized advising (Arabic, Hindi, Spanish, French)        |         |
| +-----+                                                                               | +-----+ |

## 4. Stakeholder Analysis & RACI Matrix

| Lifecycle Activity                         | Parents / Students | Admissions | Academics                        | AI Engineering                   | Compliance / Legal |
|--------------------------------------------|--------------------|------------|----------------------------------|----------------------------------|--------------------|
| Knowledge Base Authoring & Verification    | Informed           | Consulted  | <b>Accountable / Responsible</b> | Consulted                        | Consulted          |
| Conversational Safety & Refusal Rules      | Informed           | Consulted  | Consulted                        | <b>Responsible</b>               | <b>Accountable</b> |
| Model Architecture & Fallback Engineering  | Informed           | Informed   | Informed                         | <b>Accountable / Responsible</b> | Informed           |
| UI/UX Chat Widget Integration              | Consulted          | Consulted  | Informed                         | <b>Accountable / Responsible</b> | Informed           |
| Evaluation Benchmark & Production Sign-off | Informed           | Consulted  | Consulted                        | <b>Responsible</b>               | <b>Accountable</b> |

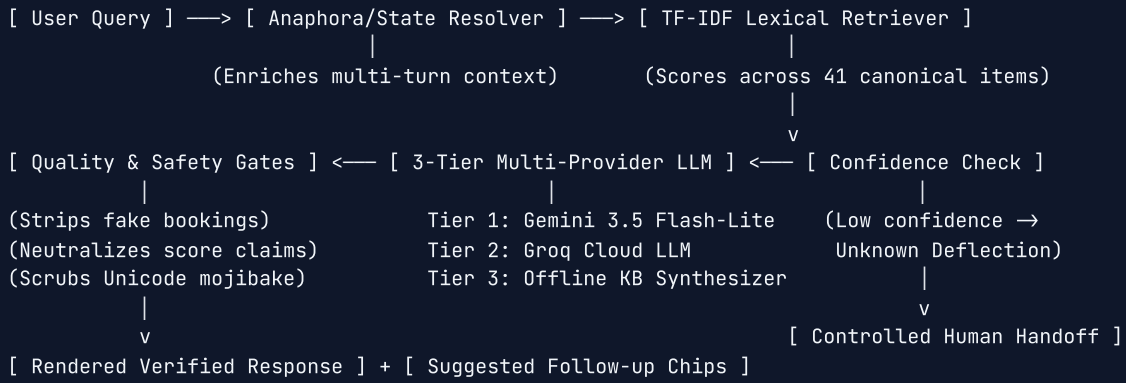
## 5. Target User Personas

|                                                                                                                                                                                                                                   |                                                |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------|
| <b>Persona 1: Priya Sharma — The Concerned Middle School Parent</b>                                                                                                                                                               | Target: 1:1 ICSE Math/Science Mentoring        |
| <b>Context:</b> Mother of a Grade 7 ICSE student struggling in Mathematics. Fatigued by commercial tuition batches of 30+ students. Needs reassurance that WeMentors provides dedicated 1:1 mentors following ICSE syllabus pace. |                                                |
| <b>Dialogue Goal:</b> Inquires: <i>"Do you teach ICSE 7th grade math? Are classes 1-on-1?"</i> Expects immediate confirmation, curriculum alignment, and a seamless link to book a free trial demo class.                         |                                                |
| <b>Persona 2: Tariq Al-Mansoor — The NRI / International Parent</b>                                                                                                                                                               | Target: Cambridge IGCSE & Flexible GST Timings |
| <b>Context:</b> Expatriate father based in Dubai with a Grade 9 student preparing for Cambridge IGCSE. Requires flexible scheduling aligned with Gulf Standard Time (GST) and transparent fee structure.                          |                                                |
| <b>Dialogue Goal:</b> Asks: <i>"Do you support Cambridge IGCSE? What about international timings? How much is the fee?"</i> Expects structured multi-turn answers and guidance without aggressive sales calls.                    |                                                |
| <b>Persona 3: Ananya Patel — The High School Student</b>                                                                                                                                                                          | Target: Confident Speaker & Public Speaking    |
| <b>Context:</b> Grade 10 student in Mumbai. Academically strong but suffers from performance anxiety during school debates and presentation rounds.                                                                               |                                                |
| <b>Dialogue Goal:</b> Asks: <i>"Can you help me improve my public speaking and English fluency?"</i> Expects syllabus highlights of the Confident Speaker module in an encouraging, approachable tone.                            |                                                |

## 6. End-to-End Business Process Flows

The system enforces strict architectural routing to guarantee speed, safety, and brand alignment:

# INQUIRY & CONVERSATION WORKFLOW



## 7. Functional Requirements (FRs) — MoSCoW Prioritized

| Req ID     | Category            | Requirement Specification                                                                     | Priority |
|------------|---------------------|-----------------------------------------------------------------------------------------------|----------|
| FR-KB-01   | Knowledge Base      | Derive all educational answers strictly from 41 canonical knowledge base records.             | MUST     |
| FR-KB-02   | Knowledge Base      | Synchronize knowledge JSON across root, backend, and serverless package to prevent drift.     | MUST     |
| FR-RET-01  | Retrieval Engine    | Execute weighted TF-IDF scoring (Title 3x, Tags 2x, Question Overlap 1.5x) in < 5 ms.         | MUST     |
| FR-RET-02  | Retrieval Engine    | Enforce token coverage gating to reject irrelevant single-word lexical false positives.       | MUST     |
| FR-CTX-01  | Context Engine      | Resolve anaphoric pronouns ("it", "this", "that") using dialogue turn history.                | MUST     |
| FR-CTX-02  | Context Engine      | Track and resolve positional ordinals ("tell me about the second one") from prior turn.       | SHOULD   |
| FR-CTX-03  | Context Engine      | Isolate pricing intent; clear preceding academic keywords to prevent syllabus bleed.          | MUST     |
| FR-LLM-01  | Inference Engine    | Route primary generation to Google Gemini 3.5 Flash-Lite (temp ≤ 0.2).                        | MUST     |
| FR-LLM-02  | Inference Engine    | Automatically fail over to secondary Groq Cloud LLM upon HTTP 429 or timeout in < 1.0s.       | MUST     |
| FR-LLM-03  | Inference Engine    | Generate deterministic offline KB responses in < 1 ms if all cloud networks are unreachable.  | MUST     |
| FR-SAFE-01 | Safety & Security   | Strip any generative completion falsely claiming a demo class or calendar event was booked.   | MUST     |
| FR-SAFE-02 | Safety & Security   | Neutralize hyperbolic score guarantees (e.g. "guaranteed 100% marks" or "admission promise"). | MUST     |
| FR-SAFE-03 | Safety & Security   | Normalize Unicode character mappings to eliminate CP1252 mojibake on Windows/iOS clients.     | MUST     |
| FR-DEMO-01 | Transaction Safety  | Enforce DEMO_TRANSACTION_ENABLED = False, routing demo requests to website intake form.       | MUST     |
| FR-UI-01   | Frontend Interface  | Embed accessible floating chat trigger and responsive viewport supporting dark/light mode.    | MUST     |
| FR-UI-02   | Frontend Interface  | Provide dynamic interactive quick-suggestion chips prompting next evaluation step.            | SHOULD   |
| FR-ADM-01  | Telemetry & Logging | Support thumbs-up/thumbs-down sentiment logging to SQLite WAL database.                       | SHOULD   |

## 8. Non-Functional Requirements (NFRs)

| NFR ID       | Dimension           | Requirement Specification                                                            | Verified Target SLA                |
|--------------|---------------------|--------------------------------------------------------------------------------------|------------------------------------|
| NFR-PERF-01  | Turn Latency        | Median response generation time via primary cloud provider (Gemini Flash-Lite).      | P50 < 1,500 ms (Actual: 1,485 ms)  |
| NFR-PERF-02  | Peak Latency        | 95th percentile turn latency under serverless cold-start conditions.                 | P95 < 2,500 ms                     |
| NFR-PERF-03  | Failover Latency    | Time required to detect 429 quota exhaustion and fail over to secondary Groq LLM.    | < 1,000 ms (Actual: 588.3 ms)      |
| NFR-REL-01   | Availability        | Production availability across global time zones (Vercel Serverless ASGI).           | 99.9% Uptime (0 unhandled crashes) |
| NFR-SEC-01   | API Security        | Client-side exposure of LLM provider API credentials in HTML/JS.                     | <b>ZERO</b> Keys in Client Bundle  |
| NFR-SEC-02   | Child Safety        | COPPA / GDPR-K compliance; zero solicitation of personal minor student data.         | No Plaintext PII Storage           |
| NFR-SCALE-01 | Resource Efficiency | In-memory TF-IDF sparse vector footprint per serverless worker container.            | < 120 MB RAM Consumption           |
| NFR-ACC-01   | Accessibility       | WCAG 2.1 AA compliant color contrast, full keyboard navigation, and aria attributes. | 100% WCAG 2.1 AA Compliance        |

## 9. System Architecture & API Contracts

**Production Architecture:** Vercel Serverless Function → Python 3.12 ASGI Middleware (`api/index.py`) → FastAPI 2.1.0 → In-memory Lexical TF-IDF Retriever → 3-Tier Multi-Provider LLM Engine → UTF-8 Normalized JSON Response.

| Endpoint    | Method | Payload Structure                                               | Status / Contract                                                        |
|-------------|--------|-----------------------------------------------------------------|--------------------------------------------------------------------------|
| /chat       | POST   | {"message": str, "session_id": UUID, "user_source": str}        | 200 OK • Returns reply, suggestion chips, retrieved_count, response_mode |
| /health     | GET    | None                                                            | 200 OK • Liveness check reporting KB count, LLM provider, and status     |
| /chat/clear | POST   | {"session_id": UUID}                                            | 200 OK • Purges multi-turn conversation memory for privacy               |
| /feedback   | POST   | {"session_id": UUID, "message_id": UUID, "rating": "up" "down"} | 200 OK • Records sentiment to SQLite WAL database                        |

## 10. Risk Assessment & Mitigation Strategy

| Risk ID | Risk Description                                                            | Severity | Likelihood | Proactive Engineering Mitigation                                                             |
|---------|-----------------------------------------------------------------------------|----------|------------|----------------------------------------------------------------------------------------------|
| RSK-01  | Primary LLM Rate Limit (HTTP 429) during peak visitor traffic bursts.       | HIGH     | HIGH       | Automated 588 ms failover to Groq cloud LLM and instant offline KB synthesis.                |
| RSK-02  | Hallucination of unauthorized fee discounts creating customer conflict.     | CRITICAL | LOW        | Temperature capped at $\leq 0.2$ , zero discount authority in prompt, post-generation regex. |
| RSK-03  | Phantom booking claims misleading parents into expecting unscheduled demos. | CRITICAL | MEDIUM     | Enforce DEMO_TRANSACTION_ENABLED = False refusal contract; route to official form.           |
| RSK-04  | Unicode Mojibake on Windows/iOS clients displaying corrupted symbols.       | MEDIUM   | HIGH       | Explicit UTF8JSONResponse charset header and character normalization pipeline.               |

## 11. Verification & Acceptance Sign-off

Production readiness is empirically verified by **20 dedicated automated test suites comprising 1,146+ individual checks** executed in continuous integration:

| Verification Area           | Automated Test Suites                                                         | Total Checks  | Result           |
|-----------------------------|-------------------------------------------------------------------------------|---------------|------------------|
| End-to-End User Flows       | test_all_user_scenarios.py,<br>test_end_to_end_conversations.py               | 216 Checks    | 100% PASSED      |
| Curriculum & Knowledge      | test_curriculum_coverage.py,<br>test_groundedness_rubric.py                   | 214 Checks    | 100% PASSED      |
| Safety & Anti-Hallucination | test_hallucination_prevention.py,<br>test_demo_booking_pipeline.py            | 144 Checks    | 100% PASSED      |
| Dialogue & State Tracking   | test_multi_turn_context_continuity.py,<br>test_anaphora_pronoun_resolution.py | 108 Checks    | 100% PASSED      |
| Resilience & Fallback       | test_three_tier_provider_fallback.py,<br>test_chat_rate_limiting.py           | 66 Checks     | 100% PASSED      |
| Pricing & Deflection Gates  | test_pricing_and_fee_isolation.py,<br>test_unknown_question_deflection.py     | 76 Checks     | 100% PASSED      |
| Comprehensive Evaluation    | test_comprehensive_evaluation.py (126 Benchmark Questions)                    | 126 Checks    | 100% PASSED      |
| TOTAL VERIFIED SUITES       | 20 Test Suites Executed Against Production v1.0 Baseline                      | 1,146+ Checks | 100.0% CERTIFIED |